

2025

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Y24

Booklet No.

Electrical Engineering Paper-II

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional

QUESTION PAPER SPECIFIC INSTRUCTIONS

Please read each of the following instructions carefully before attempting questions :

1. There are **EIGHT** questions divided in two Sections, out of which **FIVE** are to be attempted.
2. Question no. 1 and 5 are compulsory. Out of the remaining questions, **THREE** are to be attempted choosing at least **ONE** question from each Sections.
3. The number of marks carried by a question/sub question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note - Candidates will be allowed to use Scientific (Non-programmable type) calculators.

SEAL

SECTION – A

- 1. Answer any five of the following :** **(8x5=40)**
- (a) In relation to Op-Amp define : **(4x2=8)**
- (i) CMRR
 - (ii) SVRR
 - (iii) Output short circuit current
 - (iv) Slew rate
- (b) With the help of neat diagram explain the working of full wave rectifier using Op-Amp, derive the expression for output voltage. **(8)**
- (c) A tuned RLC circuit resonates at 100 MHz by the capacitance of 20 pF. The Q factor of the circuit is 40. If this circuit is maintained at 17°C with the bandwidth 10 kHz, calculate the effective noise voltage. **(8)**
- (d) For the quadratic equation $x^2 + 12x + 37$ having r as a base of number system. If one of the root is 8, what will be the value of r (base of the number system in which equation is written). **(8)**
- (e) Find the Nyquist rate and Nyquist interval for the following signal : **(8)**
- $$f(t) = \frac{1}{2\pi} \cos(4000 \pi t) \cdot \cos(1000 \pi t)$$
- (f) Regarding SCR define the following terms : **(4x2=8)**
- (i) Latching current
 - (ii) Holding current
 - (iii) Peak Repetitive Reverse Voltage
 - (iv) Peak Repetitive Forward Blocking Voltage
- (g) Explain operation of Bridge type 1 – ϕ voltage source inverter with R load. (circuit diagram, output expression and output waveform is expected) **(8)**
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- 2. (a) Explain the working of antilog amplifier (using Op-Amp) with circuit diagram and proof for output voltage.** **(5+5+5=15)**
- (b) An amplifier is fed from a 100 Ω , 15 μ V rms sinusoidal signal source. Its equivalent input noise resistance and equivalent input shot noise current are 250 Ω and 6 μ A, respectively. Calculate the individual noise voltage at the input and the input-to-noise ratio. Assume noise bandwidth is 10 MHz, and temperature is 30°C. **(15)**
- (c) Implement the Boolean function. $F(A, B, C, D) = \Sigma(1, 3, 4, 11, 12, 13, 14, 15)$ using 8 : 1 MUX; encode the LSB variable. **(10)**
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- 3. (a) Explain following three quantizer with its working, block diagram output (if any)** **(15)**
- (i) Counting quantizer
 - (ii) Serial quantizer
 - (iii) Parallel quantizer
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- (b) Explain SCR commutation using following three methods (separately) using circuit diagram and waveforms. (15)
- (i) Capacitor commutation
 - (ii) Commutation by resonance
 - (iii) A.C. line commutation
- (c) Explain the working of Buck-Boost Chopper with circuit diagram and waveforms. (10)
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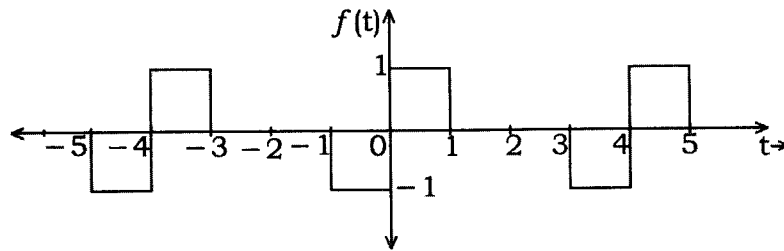
4. (a) (i) Write program in C language to add all digits of given number to get output in single digit. (8)
(eg. 14792 addition of all digit is 5)
- (ii) Calculate the value of the LSB, MSB and full scale output for an 8 bit DAC (Digital to Analog Converter) for the 0 to 10 V range. (7)
- (b) With the help of circuit diagram and current voltage diagram/waveform. Explain the working of 3 phase bridge inverter. (15)
- (c) Explain five characteristics (most important) of Radio Receiver with its Relevance, diagram, in order to Receive the signal at receiver with good SNR. (10)
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SECTION - B

5. Answer **any five** of the following : (8x5=40)
- (a) A 230 V separately excited dc motor takes 50 A at a speed 800 rpm. It has armature resistance of 0.4 Ω . This motor is controlled by a chopper with an input voltage of 230 V and frequency of 500 Hz. Calculate the duty ratio for 1.2 times rated torque and speed of 500 rpm. (8)
- (b) Obtain the initial and final values of the function whose Laplace transform is given by (8)

$$G(s) = \frac{3s^3 + 2s + 10}{s(s+2)^2(s+3)}$$

- (c) Find the Fourier series expression of $f(t)$ given in figure below. (8)



- (d) Consider $Y(k) - Y(k-1) + 0.24Y(k-2) = x(k) + x(k-1)$ where $Y(k) = 0$ for $k < 0$ (8)

Determine the weighing sequence and $Y(k)$ when $x(k)$ is unit step sequence.

- (e) If a system with impulse response $\{h(n)\} = \{1, 2, 3\}$ is subjected to an input sequence $\{u(n)\} = \{4, 5, 6\}$ (8)

Determine the output sequence $Y(k)$ by convolution. It can be assumed that both h and u start at $n = 0$ and are zero for $n < 0$.

- (f) For a unity feedback system having open loop transfer function as : (8)

$$G(s) = \frac{K(s+2)}{s^2(s^2 + 7s + 12)}$$

Determine (i) Type number and order of the system (ii) Error constants, and (iii) steady state error for parabolic input.

- (g) What are the advantages of designing compensator in frequency domain over time domain. (8)

Explain in brief the design steps of a lag compensator in frequency domain using Bode plot.

6. (a) A 400 V, 50 Hz, 4 Pole, 1370 rpm, star connected induction motor is supplied from a current regulated PWM voltage source inverter and is operated with rotor flux oriented vector control. (15)

The motor parameters are given as follows :

$$R_s = 2 \Omega, R'_r = 5 \Omega, X_{ls} = X'_{lr} = 5 \Omega, X_m = 80 \Omega$$

All reactances are calculated at 50 Hz. Neglect core and friction losses

- Draw the steady state per phase equivalent circuit diagram with labels.
- Determine the required value of d axis stator current ' i_{ds} ' and q axis stator current ' i_{qs} ' to operate the motor at rated speed if the terminal voltage and freq. are held at rated value.

- (b) Determine the solution of following differential equation using Laplace transform method. **(15)**

$$\frac{d^2V(t)}{dt^2} + 6\frac{dV(t)}{dt} + 8V(t) = 2u(t) \text{ subject to } V(0) = 1, \frac{dV(0)}{dt} = -2$$

- (c) Consider the following difference equation **(10)**

$$Y(k+2) + 0.5Y(k+1) + 0.06Y(k) = -(0.5)^{k+1}$$

with initial conditions $Y(0) = 0, Y(1) = 0$

Find the solution $Y(k)$ for $k > 0$.

7. (a) Define gain margin, phase margin and bandwidth related to the stability of the system in frequency domain. **(15)**

The open loop transfer function of a unity feedback control system is given by :

$$G(s) = \frac{4}{s(s+1)}$$

Determine the gain margin, phase margin, resonant peak and bandwidth of the closed loop system.

- (b) The open loop transfer function of a unity feedback control system is given by **(15)**

$$G(s) = \frac{2(1-s)}{(s+2)(s+3)}$$

Sketch the Nyquist plot (roughly) and use the Nyquist criterion to determine whether the closed loop system is stable.

- (c) State the properties of discrete Fourier transform. **(10)**

Compute the 2 point discrete Fourier transform of the following sequence.

$$x[n] = \{1, 2\}$$

8. (a) In the case of speed control of Induction motor using V/f control, why it is necessary to maintain the voltage to frequency ratio constant ? Explain with block diagram a closed loop induction motor drive with constant V/f control strategy. (15)
- (b) Consider a third order system with open loop transfer function. (15)

$$G(s) = \frac{1}{(0.1s+1)(0.2s+1)^2}$$

Design a PID controller based on Ziegler – Nichols method to achieve fast response, minimal oscillation and stable operation.

Use Routh-Hurwitz stability criterion to determine critical period and critical gain.

- (c) Consider the system defined by $\dot{X} = AX + Bu$ (10)

Where,

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -5 & -6 \end{bmatrix}; B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

By using the state feedback control $u = -KX$ it is desired to have the closed loop poles at $S = -2 \pm j4$, $S = -10$. Determine the state feedback gain matrix K .

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